

Week 2 Report

Running Code Llama on the Virtual Machine

Objective

Install and run a local LLM (Code Llama) on the Virtual Machine using Ollama, and verify it responds correctly to prompts.

What Was Done

- Installed [Ollama](#) on the VM — a tool for running open-source LLMs locally.
- Pulled the **Code Llama 7B Instruct** model (`code llama:7b`), a ~3.8 GB download.
- Configured the VM with sufficient resources (upgraded RAM from 4 GB to 8 GB) after observing that the default allocation was too small to comfortably run a 7-billion-parameter model.
- Verified the model with a live prompt.

Verification

Prompt sent:

```
Write a Python function to check if a number is a palindrome.
```

Response (abridged):

```
def is_palindrome(n):  
    return str(n) == str(n)[::-1]
```

The model responded correctly and coherently, confirming Ollama + Code Llama were fully operational on the VM (CPU-only mode, since no GPU is passed through to the VM by VMware by default).

Key Learnings

Model sizing vs. hardware: a 7B-parameter model needs several GB of RAM just to load; running it on constrained hardware (CPU-only, limited RAM) is slower but functionally correct.

Instruction-tuned prompting: Code Llama's "Instruct" variant is fine-tuned to follow natural-language commands (e.g. "Write a function to...") rather than just continuing code, using the `[INST] ... [/INST]` prompt format internally.

Outcome

A working local LLM environment on the VM, reachable via Ollama's HTTP API at `http://localhost:11434`, used directly by the application built in Week 3.